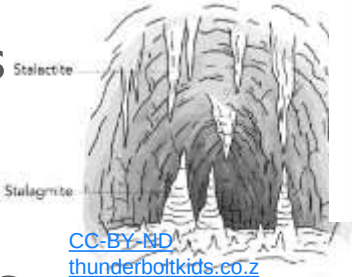
Week 1 Cyber Security Technologies

- Welcome!
- Cybersecurity includes, e.g.
  - Detecting threats, e.g. malicious insiders
    - Threat Actors (TAs) cause incidents via vulnerabilities
  - Protecting CIA, e.g. encryption
  - Managing risks, e.g. prioritise protections
    - McCumber's cube helps identify risks
- Virtual machines are guests on a host
- PowerShell, e.g. **pwd**, **ls**, **echo**, **cat**, >
  - `Object pipe | , Get-Member`
  - `(ls).Name`
  - `ls | Where-Object Name -like hello.*`
- PowerShell networking

	PS> Powershell	Deprecated
IP	`Get-NetIPAddress`	ipconfig
Test connection	`Test-NetConnection`	ping



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thunderboltkids.co.za

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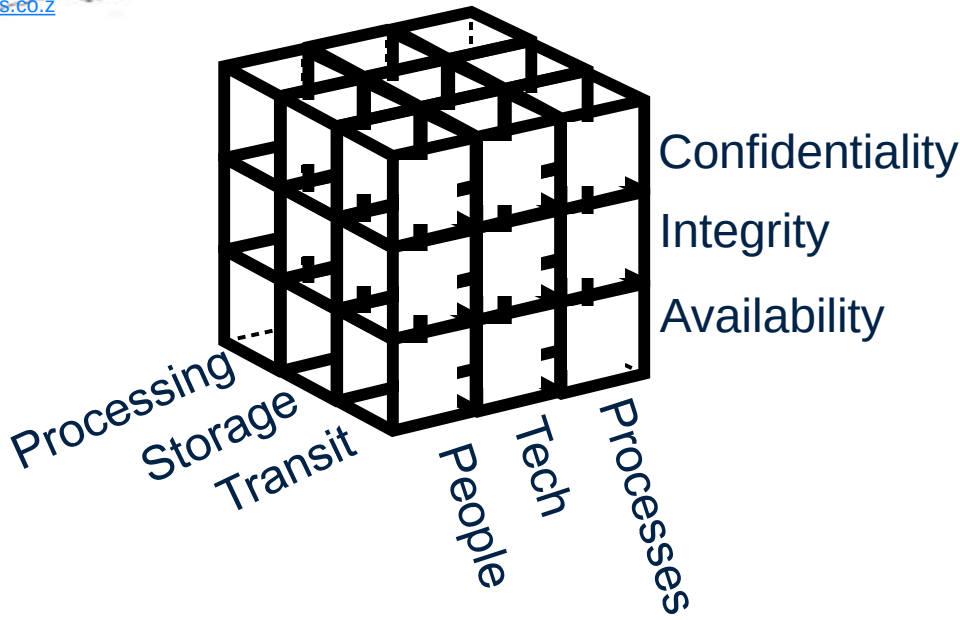
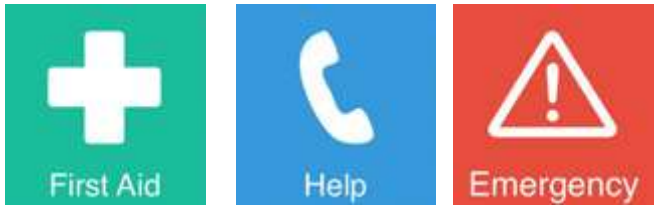


Emergency Procedures

- Evacuation procedure
- Security
  - 1331 & 0418 792 982
- SafeZone app



Unit Objectives

- Develop PowerShell, Bash & Python scripts to implement Windows & Linux defensive safeguards
- Use cybersecurity tools to identify vulnerabilities
  - “Don’t think like a castle; think like an invader”

Igor Dekhtyarchuk
FBI wanted: id theft

Support

- Forums, Tutors, Unit coordinator
- Unit drop-in session
- Academic Learning Centre
  - DuckDuckGo for alc :cqu
  - 07 49707211, alc-advice@cqu.edu.au
    - Query service with 24 hr turn-around
    - [Weekly] Appointments, e.g. help you start writing an assignment
    - Assignment review service – 3 day turn-around
  - Online & oncampus workshops
  - Referencing guides, study tools

IT Resources Required

- ≥128Gb free disk space or USB 3.0
- ≥12Gb RAM
- VirtualBox
  - Windows × 2
  - Linux × n, e.g. Logger, Att (Kali), Ms2 (Metasploitable 2)
  - OpenWrt, e.g. Router
- Office 365 $0 including Excel but not Access
  - http://cqu365.cqu.edu.au
- Webcam, microphone & speakers (or headset)
- Mac
  - Not supported but many tasks possible

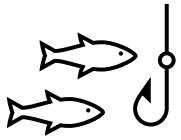
Assessment

- 3 Assessments broken into ≈ weekly tasks
  1. 3 Quizzes
     - PowerShell, Bash & End of unit review
  2. Portfolio
     - Contribute to your job interview portfolio
       - Implement safeguards on a(n) (small business) information system (IS)
       - Attack the IS, e.g. improve ability to detect
  3. Presentation
     - Demonstrate Assignment 2 safeguards & attacks

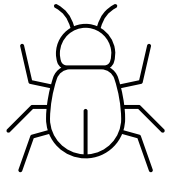
Cybersecurity

- Cybersecurity≈ manage risks to reduce info security incidents & impacts, e.g.
  - Threat of hardware failures
    - We might lose info due to a drive failure
  - Threat of trespass, e.g. physical, PCs & networks
    - Poor policies allowing unauthorised physical access to our servers
      - Rational paranoia of, e.g. painters
    - Naïve users accidentally circumvent our security technologies
      - Do family members use weak passwords?

Power loss

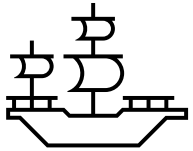


Phishing



Bugs

Obsolete tech



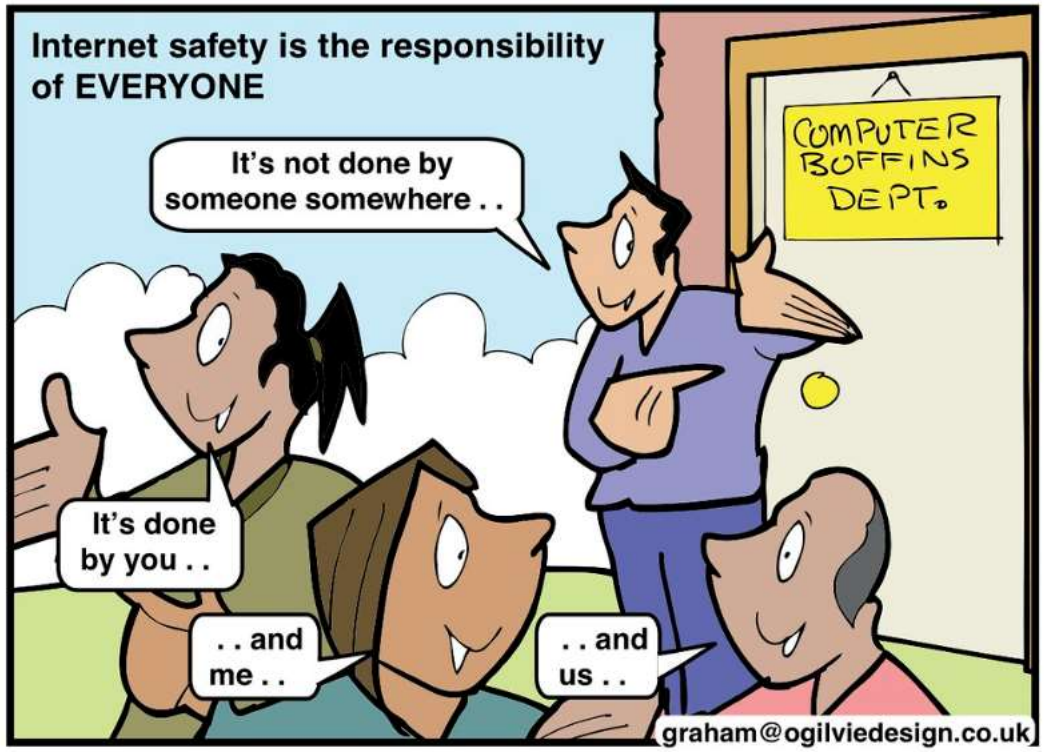
Piracy

SQL injections

Lightening

Cybersecurity is not just ICT

- Everyone in organisation responsible for cybersecurity
  - Not just ICT
  - Designers might store Intellectual Property (IP) in cloud



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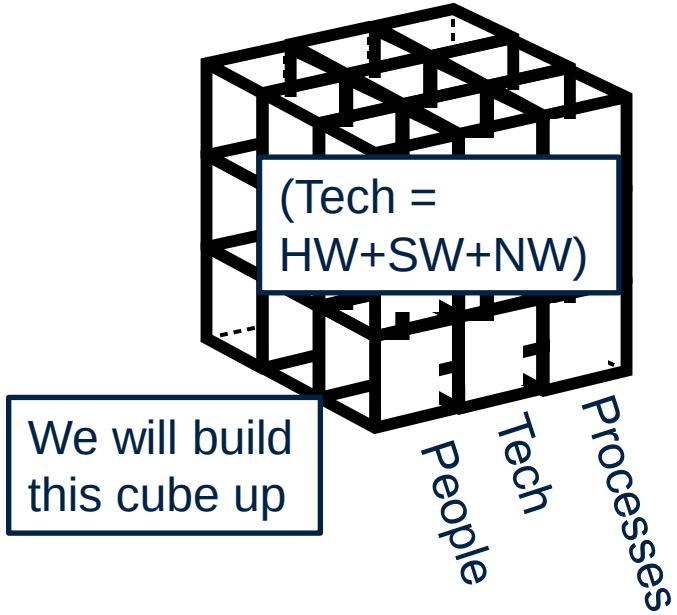
Cybersecurity Scope

- Need to consider all system components

System Component	Example Risk
Network	Battery no longer made for old UPS
Hardware	Denial of service of server with botnet
Software	SQL injection, e.g. code allowed in website search
People	Poor access control policy not enforcing multi-factor auth logins
Processes	Untrained staff who don't check tradespeople's identify
Info	Store crucial data encrypted

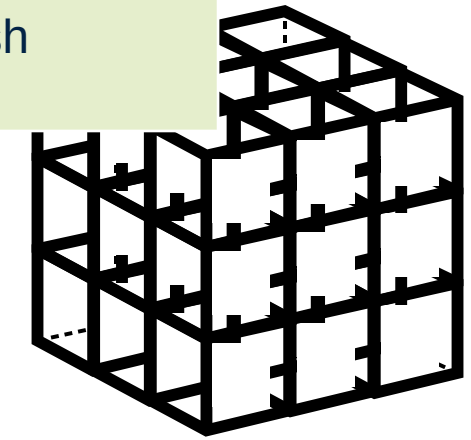


(Chp 3, Green 2015)

CIA Triad

- Need to consider each cybersecurity goal

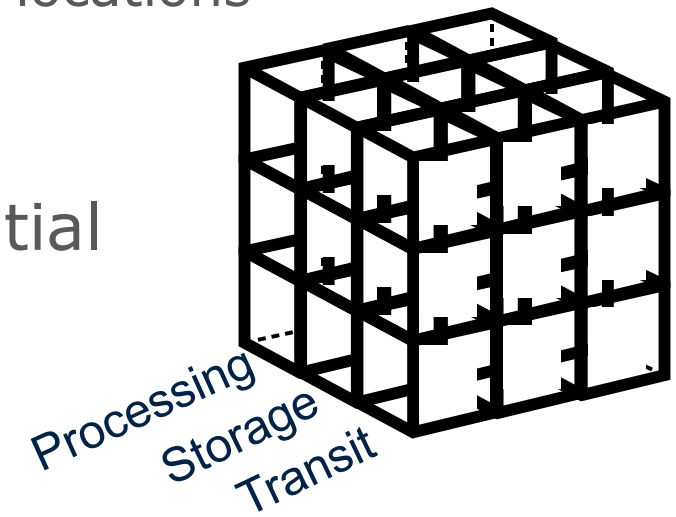
Cybersecurity Goal		Example Control / Measure
Confidentiality	Data unavailable if not needed	RAID
Integrity	Data not altered accidentally or deliberately	Encryption
Availability	Data accessible when needed	MD5 hash



(Calder 2020; Neil 2020 Chp 1)

Data States

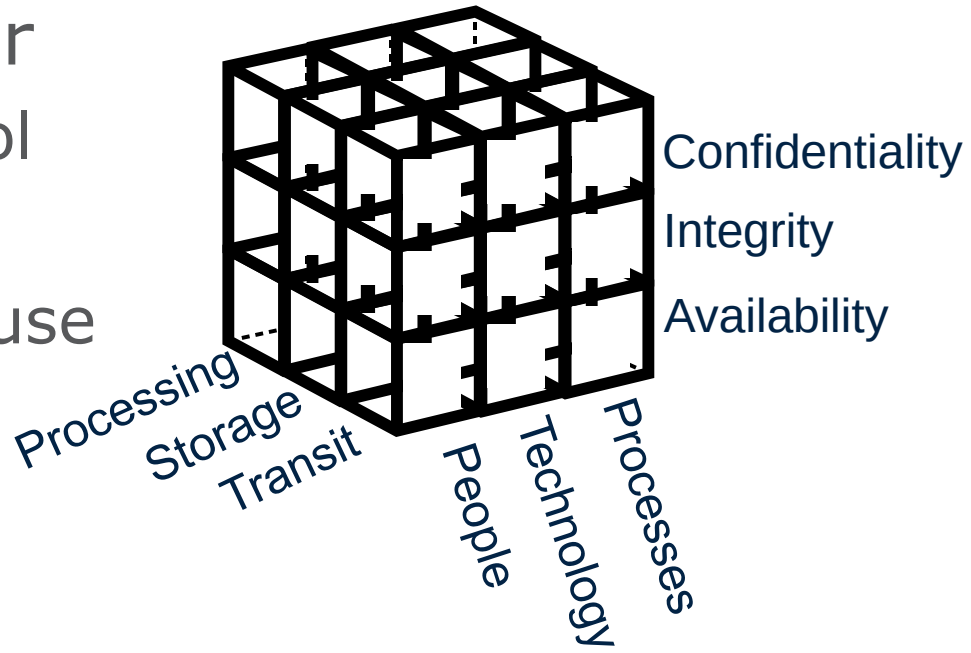
- Consider data states to identify riskiest data
  - Location
    - More environments (e.g. cloud) increase attack surface
    - Network endpoints most vulnerable
      - Where most valuable data is least protected, e.g. PCs & mobiles
  - Types
    - Storage, e.g. consider partitioning data across multiple locations
    - Processing: more vulnerable than stored data
    - Transit: most vulnerable
  - Classification, e.g. public, proprietary & confidential
    - Classify all your data



(Thurlbeck 2018) No longer available: *Cyber threat prevention*.

McCumber Cube

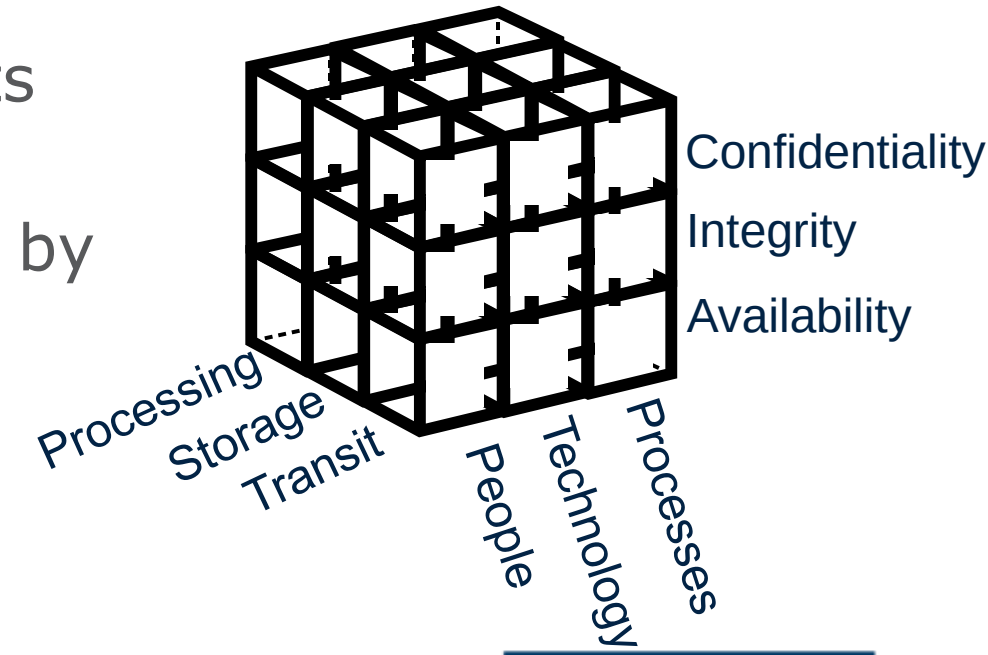
- McCumber ≈ Goals × Info States × Security Measures
- Use as checklist, e.g. for confidentiality consider info in storage, in transit & when processing
- E.g. for confidential storage consider
  - Technology: choose best encryption tool
  - People: teach to use tools encryption
  - Processes: set policies for appropriate use of info when unencrypted



System components
(Tech = HW+SW+NW)

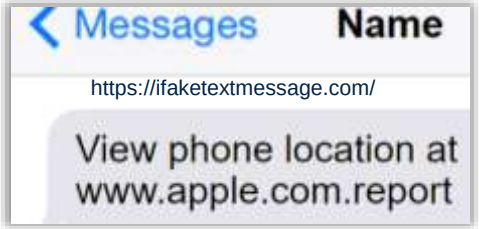
McCumber Cube

- Use McCumber Cube as checklist
- E.g. for integrity during processing consider
  - Technology: data validation applied to credit card numbers
  - People: teach how record locking affects usability
  - Processes: large transactions approved by supervisor



Failure Based Threat Model

- STRIDE-LM is a list of breaches
  - Threats to your bank account data



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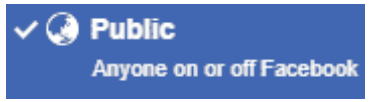
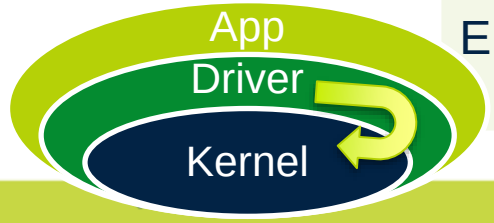
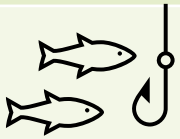
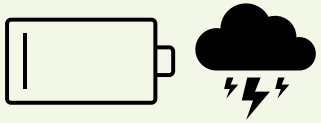
Digital signature





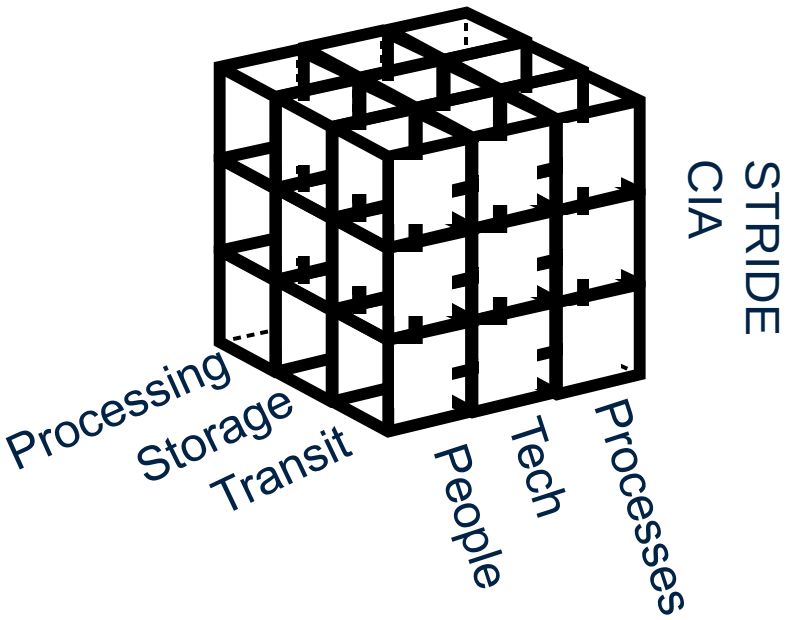
Spoofing CC SA 3.0 Unported BY The Photographersehrlich	Harddrive corruption alters your bank balance
Tampering -r-x-	Phishing email impersonating bank requesting password change 
Repudiation	Attacker denies they opened a credit card in your name
Info disclosure --wx-	Bank's server stop responding, e.g. DDoS, lightening or power loss 
Denial of Service	Attacker to packet-sniff your password when you sync browser's saved passwords
Elevation of privilege	Attacker uses SQL injection in bank's web app to change from guest to 'become' you

LM = Lateral movement

Threat Model

- STRIDE-LM ≈ attack perspective of CIA / CIAAAN

	Breach of
Spoofing	Confidentiality • Encryption
Tampering	Integrity • MD5 hashes
Repudiation	Availability • RAID
Info disclosure	Authenticity • Passwords/MFA
Denial of Service	Authorisation • Permissions
Elevation of privilege	Non-repudiation • Audit logs



Broader List of Info Security Threats than STRIDE-LM

Threat	Description/Examples
People errors	E.g. social engineering, phishing & server misconfigurations
Software attacks	Backdoor, DoS, Interceptions (MITM, sniffing, DNS poisoning, session hijacking, spoofing), malware
Info extortion	Ransomware, blackmail & info disclosure
Espionage & trespass	Unauthorised access, e.g. via password attacks
Theft	Illegal confiscation of assets
Technological obsolescence	Outdated tech
Forces of nature	E.g. cyclone, fire, flood & lightening
Technical hardware failures or errors	E.g. leaky capacitors
Technical software failures or errors	Bugs, code performance issues & loopholes
Changing quality of services	E.g. blackout, brownout, Spike, NBN outage
Sabotage & vandalism	Destruction of assets
Intellectual property compromises	Piracy & copyright infringement

(Table 1.1, Whitman & Mattford 2019)

Inadequate Responses to Incidents

- Lead to avoidable losses of money & reputation
- Reasons for poor responses include

Reason	Description
Unaware of risks	We don't have response processes, e.g. as we don't know the risks
No detection	Our response processes are not enacted as, e.g. no security alarm to identify a thief or no IDS to detect a network intrusion
Unaware how to manage incidents	We don't know the process to follow – lack of communication
Poor processes	E.g. we know our response process but it is impossible to follow, e.g. too technical

(Chp 16, Green 2016)

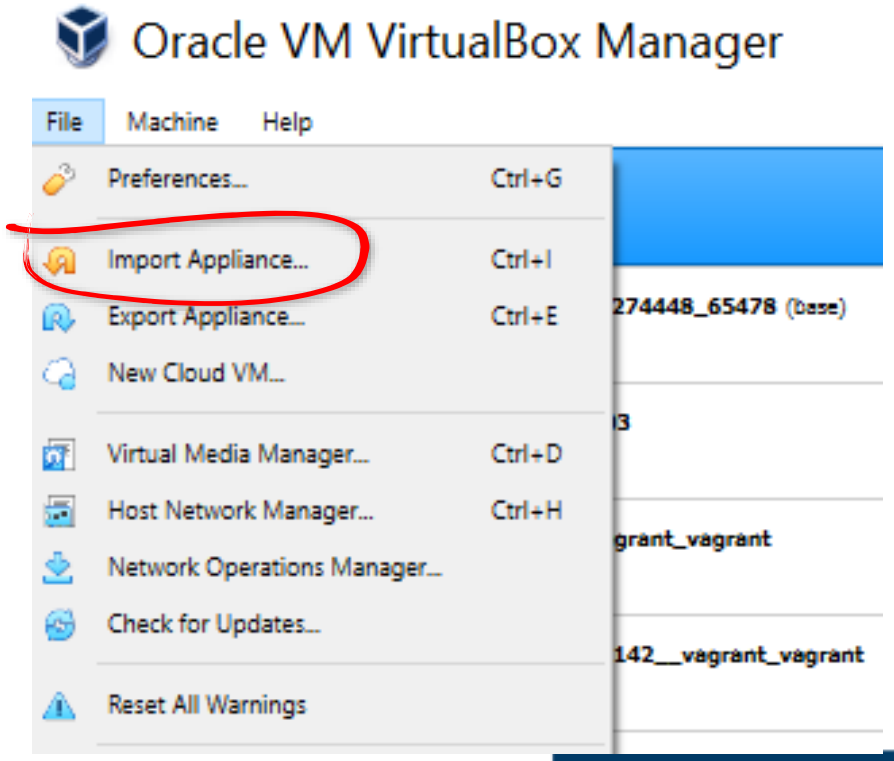
Preparing for an Incident

Process to Define	Description
Identify	Predict the riskiest scenarios, e.g. dark web purchases
Detection	E.g. setup bank notifications of withdrawals; social media reports nonresponsive website
Response	How to prevent further damage or enable quick recovery, e.g. turn off router or lease bandwidth to handle DDoS
Escalate during response	Are right people involved? When should kids raise a possible issue with mum/dad, e.g. WiFi goes down? When are lawyers needed?
Document during response	What details should be recorded? The source IP? Phisher's phone number?
Communicate during response	Who needs to be told about the incident while it is happening, e.g. • Mum/dad tell kids of router maintenance to avoid data loss • Inform customers of phishes involving our organisation
Test	Check if people's awareness & skills & technical responses are successful, e.g. have someone in a lab coat try to access offices
Recovery	How to make insurance claims; review processes; repair corrupted info

(Chp 16, Green 2016)

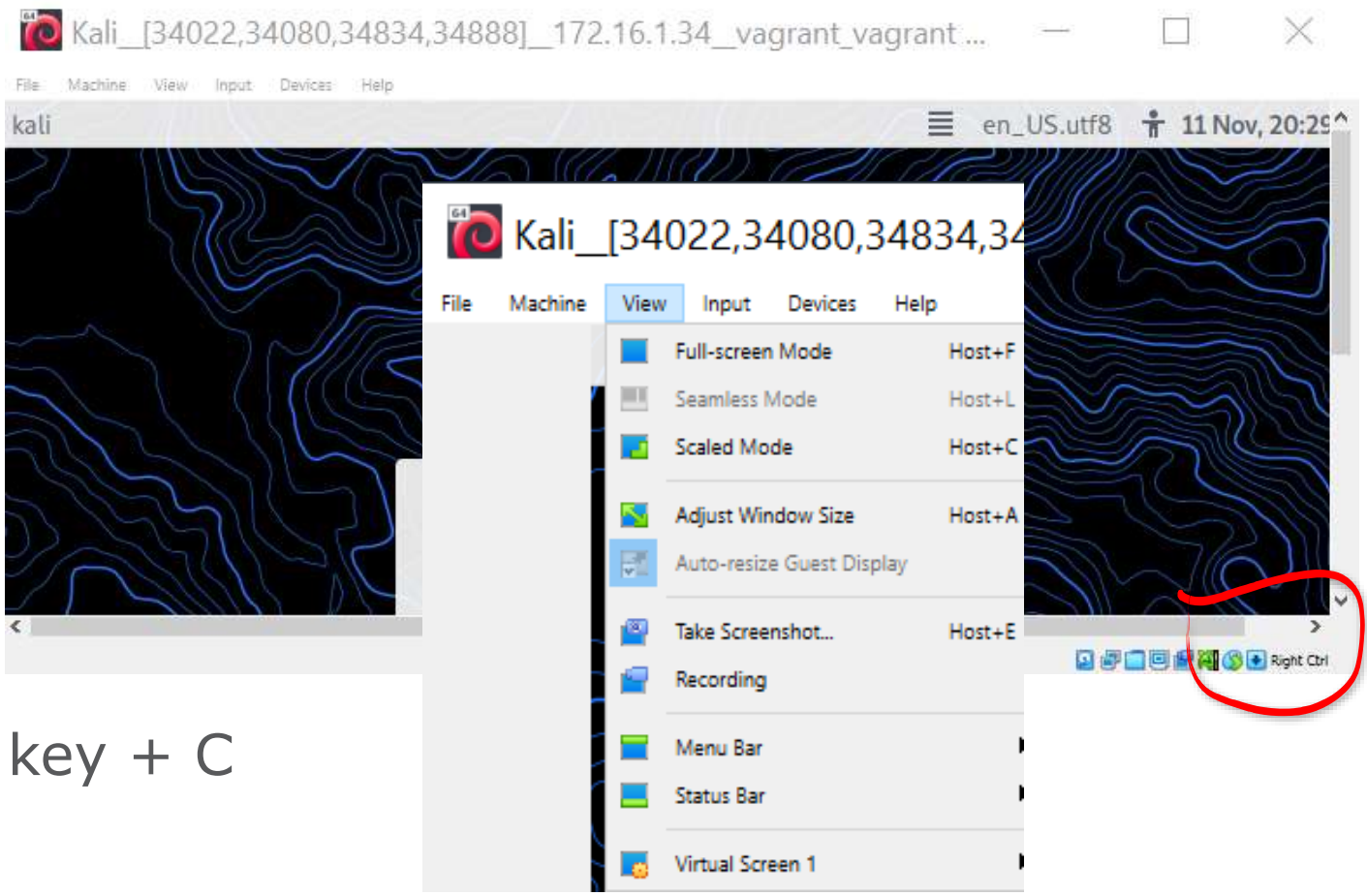
Starting a Virtual Machine with VirtualBox

- Install Oracle VM VirtualBox on your host
- Download the dc VM from unit website
- Start Oracle VM VirtualBox Manager
  - Import the Virtual Machine (VM)
    - File > Import Appliance
- Terminology
  - Host – your PC
  - Guest – VM's OS, i.e. dc



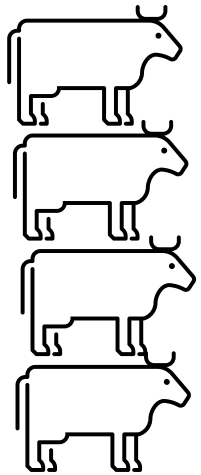
VM GUI with VirtualBox

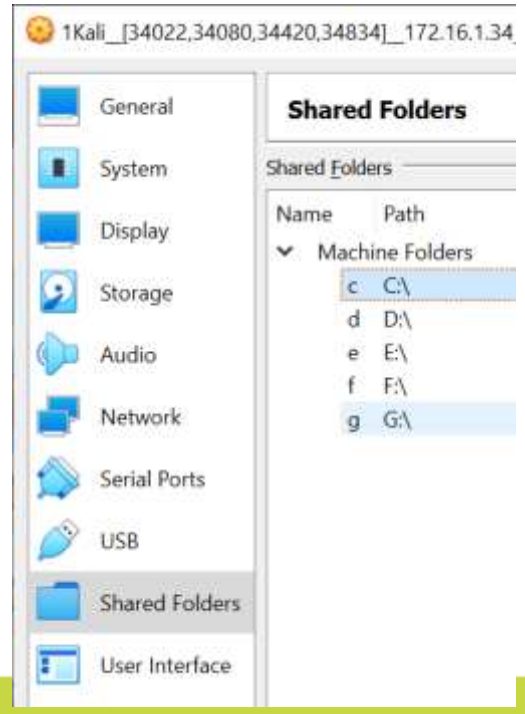
- Start the DC Guest VM
  - DC > Machine > Start
- Host key
  - Some VM GUIs need to capture your mouse
    - **Press Right Ctrl to release your mouse**
    - **Left Command** ⌘ for Mac?
- Scaled mode
  - See entire desktop
  - Exit scaled mode using Host key + C
    - Right Ctrl + C

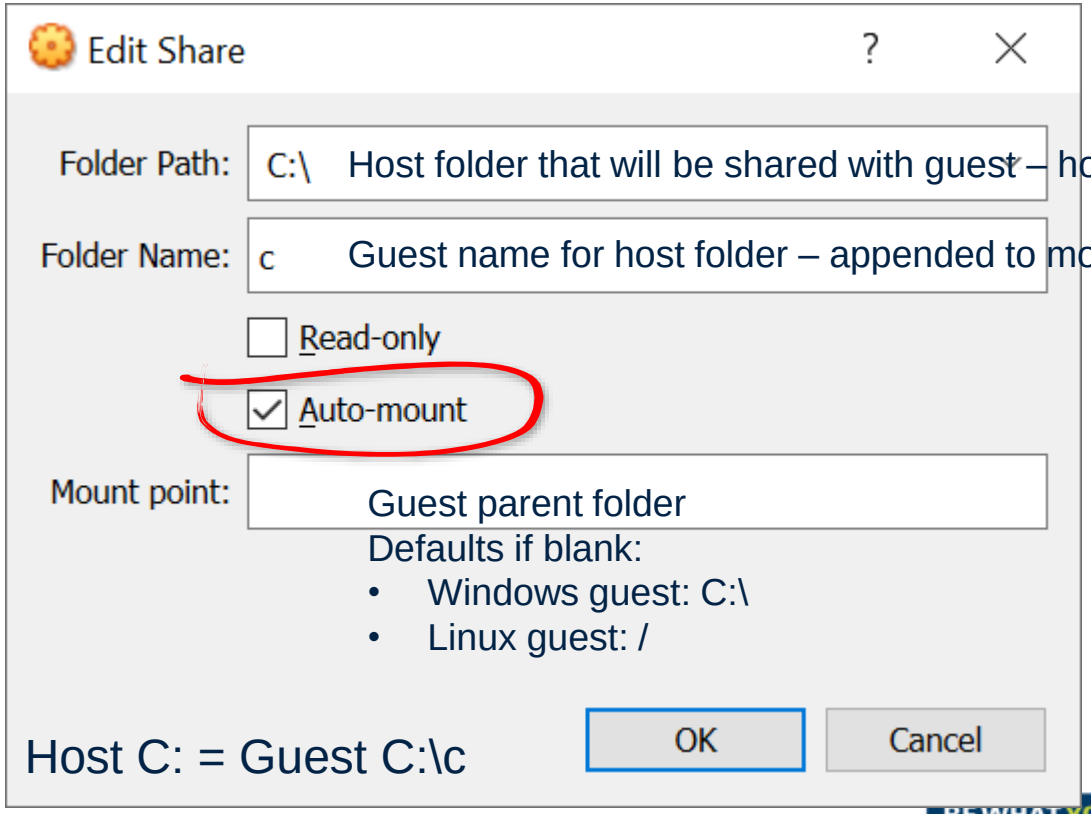


Shared Folders

- Don’t store any files in guests
- Shared folders allows you to copy files to & from host
  - Create your own shared folders for easy access
    - Use Auto-mount to provide access after VM reboot

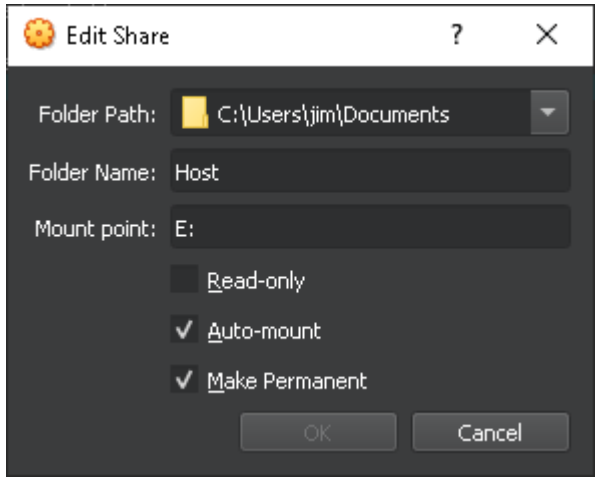




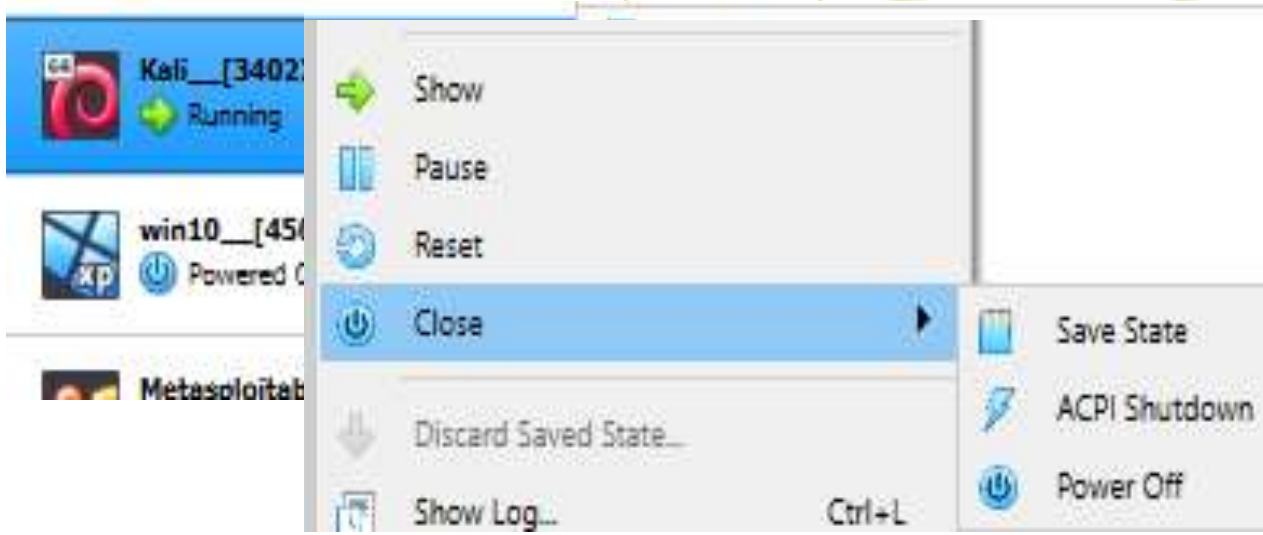


Shared Folders

- Make host C:\Users\jim\Documents accessible at guest E:\Host
  - Folder path: C:\Users\jim\Documents
  - Folder name: Host
  - Mount point: E:



VM Control in VirtualBox

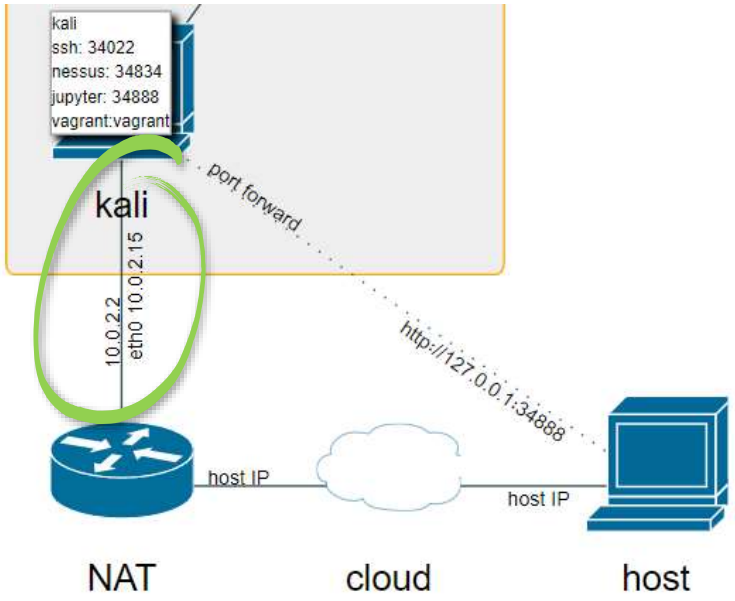


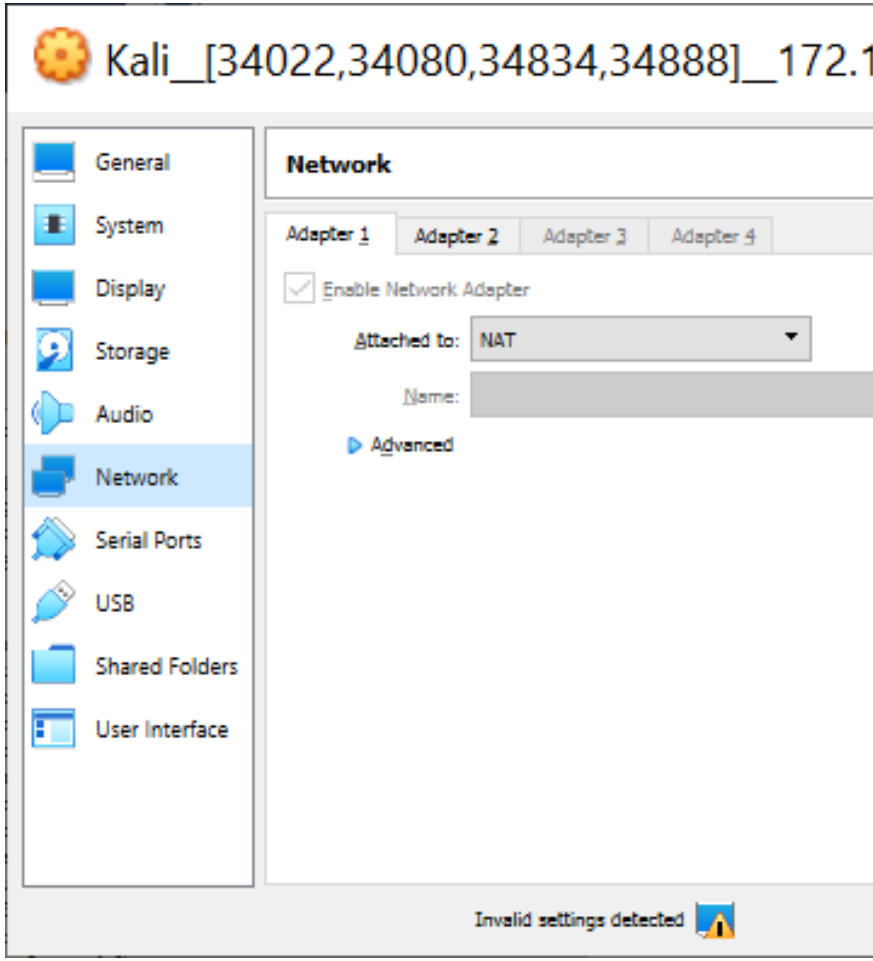
- Oracle VM menu
  - Show
    - Bring up VM terminal
  - Pause
    - Use if your host is overwhelmed
    - Sessions will be aborted if your host is shutdown & data might be lost
  - Reset: like toggling power switch
  - Close >
    - Save State: like 'hibernate'
      - Use to pause your guest when you need to shutdown your host
    - ACPI Shutdown: gracefully shutdown guest
    - Power Off : turn guest power switch off

Virtual Networking

- Settings > Network > Adapter 1
  - NAT (Network Address Translation)
    - Allows VM to access internet via host
    - Guest VMs share host IP when accessing Internet
      - Guests allocated IP, e.g. 10.0.2.15/24(?) & connected to a NAT router emulated by VirtualBox
      - Guest$_{Kali}$ ← eth0 10.0.2.15 → NAT ←$_{Host\ IP}$→ Internet

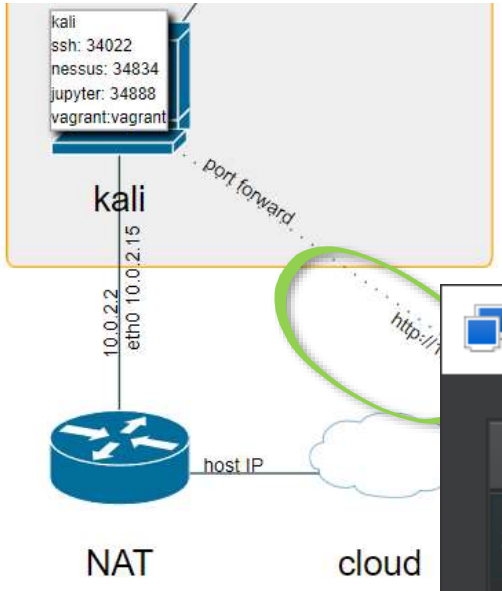
We avoid 10.0.0.0 IPs – you will see them in use though



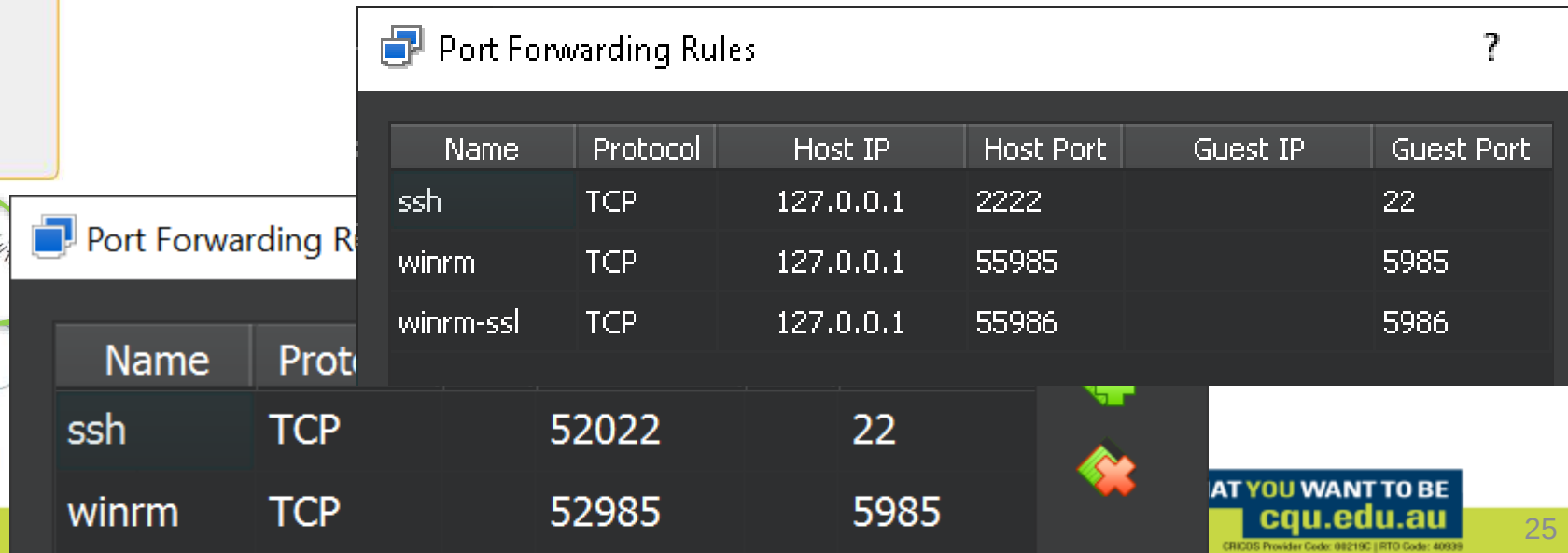


Virtual Networking

- Settings > Network > Adapter 1 (eth0)
  - Port forward allows Host to access Guest VM ports across NAT
    - Guest's wirm service accessible on host from port 55985
      - Start dc guest
      - Host PS> Test-WSMan 127.0.0.1 -port 55985
        wsmid : http://schemas.dmtf.org/wbem/wsman/identity/1/wsmanidentity.xsd
    - We avoid port forwarding
      - Use port forwarding to access guest from host if you aren't a host admin
    - Ensure host ports don't collide across guests

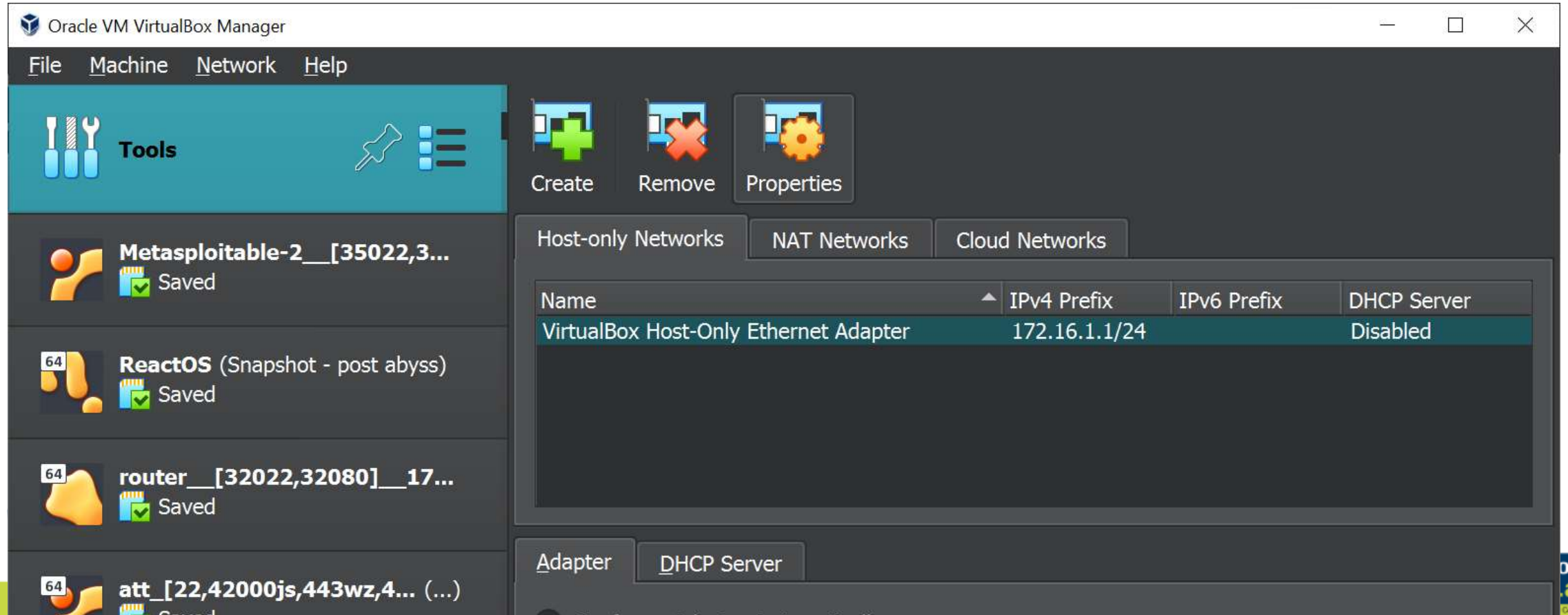


Guest Dc > Settings > Network > Adapter 1 > Advanced > Port forwarding



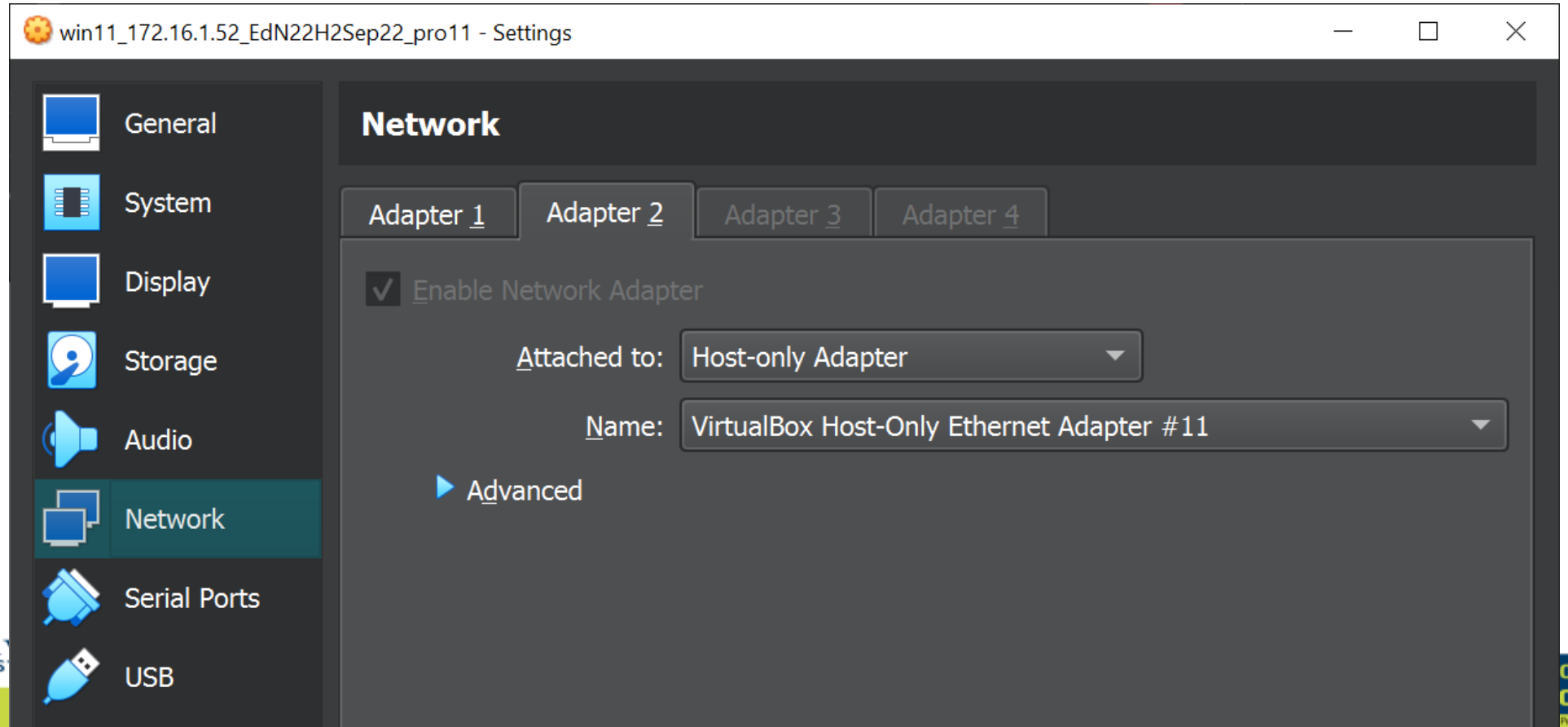
Virtual Networking

- Host-Only network adapter
  - Adapter ≈ Switch connected to host & VM guests
  - Allows host to talk to group of VM guests



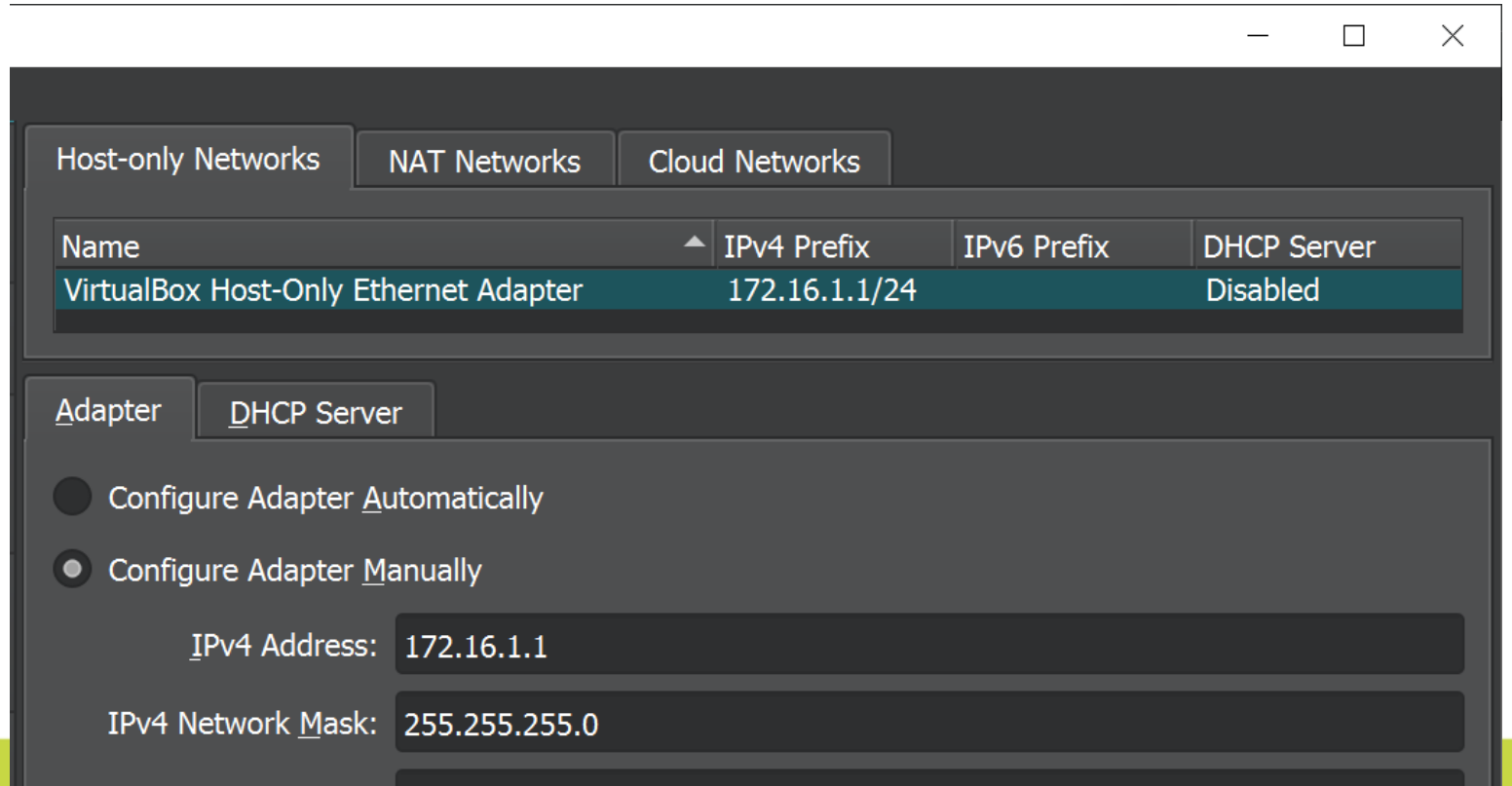
Virtual Networking

- Settings > Network > Adapter 2
  - Host-Only network adapter
    - Ensure all VMs are using same adapter, e.g. #11 - need to be host admin



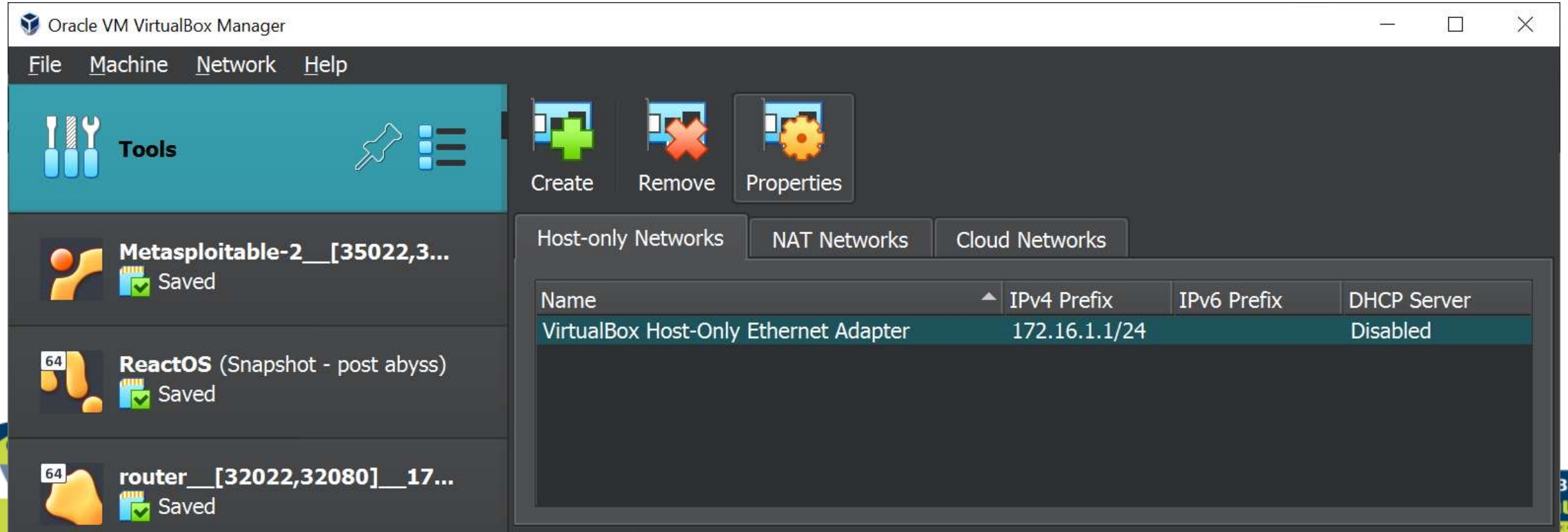
Virtual Networking

- Settings > Network > Adapter 2
  - Host-Only network adapter
    - If host admin, configure host adapter to use 172.16.1.1
      - Virtualbox > File > Tools > Network Manager
    - Otherwise, use Windows guest to connect to other guests or use port forwarding on Adapter 1



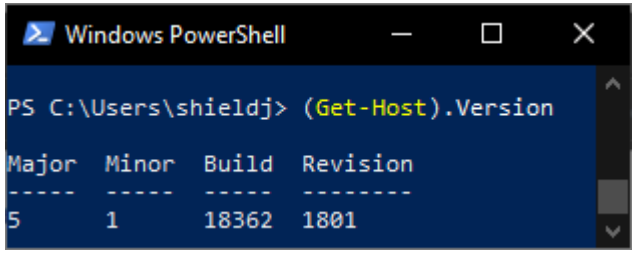
Virtual Networking

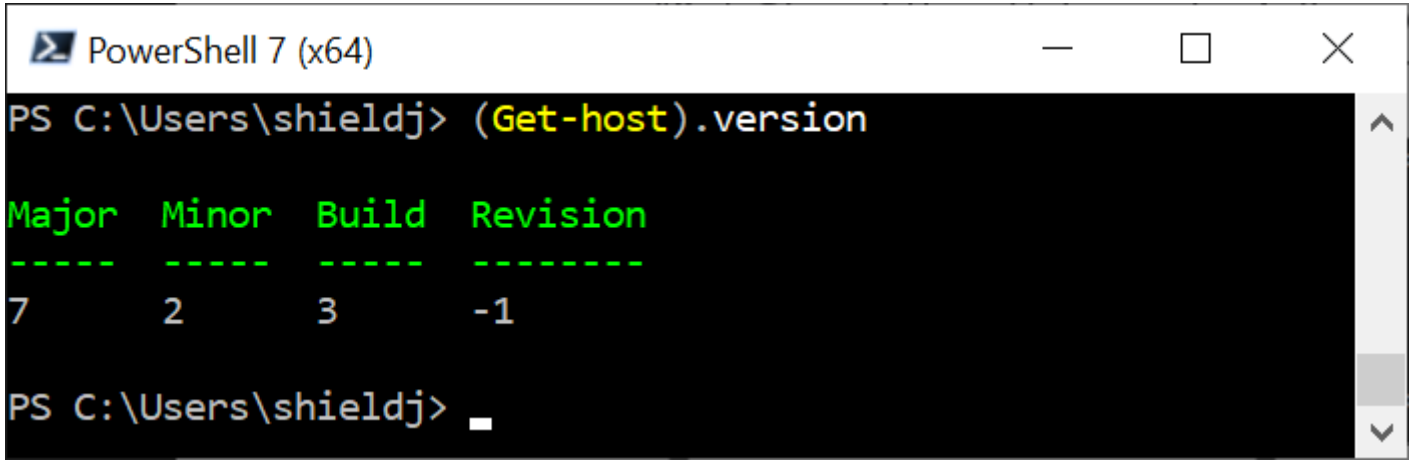
- Settings > Network > Adapter 2
  - Test Host-Only network adapter if configured to 172.16.1.1
    - Test connection to DC
      - Host PS> Test-NetConnection 172.16.1.50
        PingSucceeded : True



Start Powershell

- R > pwsh
  - If that doesn't work
    - R > powershell
    - (Get-Host).version
    - Upgrade to at least version 7
      - Chocolatey
        - » `PS> choco install pwsh`
      - PS> iex "& { $(irm https://aka.ms/install-powershell.ps1) } –UseMSI"





https://www.thomasmaurer.ch/2019/03/how-to-install-and-update-powershell-6/

PowerShell Networking

- IP – hierarchical network address of a network card
  - Used to route packets across multiple networks by grouping devices into nested
  - What are your IP address[es]?
    - PS> **Get-NetIPAddress**
      What is the CIDR (classless) address of 127.0.0.1?
    - 127.0.0.1/8
  - What is the network address?
    - 127.0.0.0/8 (replaces 0 with 0)

```
PS> Get-NetIPAddress

IPAddress         : 127.0.0.1
InterfaceIndex    : 1
InterfaceAlias    : Loopback Pseudo-Interface 1
AddressFamily     : IPv4
Type              : Unicast
PrefixLength      : 8
PrefixOrigin      : WellKnown
```

Deprecated C:> ipconfig $ ifconfig

$ ip addr

PowerShell Networking

- Test connection to a host
  - PS> **Test-NetConnection** <host>

```
PS> Test-NetConnection www.cqu.edu.au

ComputerName            : www.cqu.edu.au
RemoteAddress           : 202.9.95.188
InterfaceAlias          : Ethernet
PingSucceeded           : True
PingReplyDetails (RTT)  : 39 ms
```

Deprecated C:> ping

$ ping

PowerShell Networking

- How would you test connection to a host using HTTPS (port 443)?

```
PS> man Test-NetConnection
NAME
    Test-NetConnection

SYNTAX
    Test-NetConnection [<string>] [-TraceRoute] [-Hops <int>] …
    Test-NetConnection [<string>] [-CommonTCPPort] {HTTP | RDP | SMB | WINRM} …
    Test-NetConnection [<string>] -Port <int> …
```

PowerShell Networking

- Test connection to a host using HTTP
  - PS> **Test-NetConnection** -port 80 www.cqu.edu.au
  - PS> **Test-NetConnection** www.cqu.edu.au HTTP
- Test connection to a host using HTTPS

```
PS> Test-NetConnection -port 443 www.cqu.edu.au

ComputerName      : www.cqu.edu.au
RemoteAddress     : 202.9.95.188
RemotePort        : 443
InterfaceAlias    : Ethernet
TcpTestSucceeded  : True
```

Common Shell Commands

- Echo: outputs a string (alias for **Write-Output**)

```
PS> echo "hello"
hello
```

- File redirect: > (think of as | **Set-Content**)

```
PS> echo "hello" > hello.txt
```

- Cat: file content (alias for **Get-Content**)

```
PS> cat hello.txt
hello
```

Common Shell Commands

- Diff: compares objects (alias for **Compare-Object**)

```
PS> diff "hello" "help"
PS> diff (echo "hello") "help"

InputObject SideIndicator
help        =>
hello       <=
```

- Diff: compare the contents of hello.txt with “help”

```
PS> diff (cat .\hello.txt) "help"
InputObject SideIndicator
help        =>
hello       <=
```

Common Shell Commands

- Expect results to differ from other shells
- pwd: Print working directory (alias for **Get-Location**)

```
PS> pwd
Path
----
C:\Users\shield
```

- ls: Output each directory entry (alias for **Get-ChildItem**)

```
PS> ls
Directory: C:\Users\shield

Mode                LastWriteTime Length Name
----                ------------- ------ ----
d-----    11/03/2021 12:48PM             Documents
```

Object Pipeline

- Pipe (|) metaphor: pass stream of objects between commands
  - Ls generates an object for each file & subdirectory in directory
  - Each object contains info, e.g. attributes (e.g. read only) & extension (.txt)
  - Get-Member displays properties & methods of pipeline objects

```
PS> ls | Get-Member
    TypeName: System.IO.DirectoryInfo
...

    TypeName: System.IO.FileInfo
Name                        MemberType     Definition
----                        ----------     ----------
Extension                   Property       string Extension {get;}
Length                      Property       long Length {get;}
Name                        Property       string Name {get;}
...
```

  - What is the name of the method used to move files? MoveTo

Modify Pipeline using Dot operator

- Use Dot operator to select property
  - ls streams multiple objects
    - Dot operator is applied to each object separately

```
PS> (ls).Name
 Documents
 hello.txt
```

Modify Pipeline using Dot operator

- Get-NetAdapter shows NIC properties

```
PS> Get-NetAdapter

Name                                    Status        LinkSpeed
----                                    ------        ---------
vEthernet (Default Switch)              Up            10 Gbps
```

- Output status of your NIC, i.e. Up

```
PS> (Get-NetAdapter "vEthernet*").Status
Up
```

Filter Pipeline using Where-Object

- Filter pipeline Where-Object

```
PS> ls | Where-Object Name -like hello.*

    Directory: C:\Users\shield
Mode                LastWriteTime Length Name
----                ------------- ------ ----
-a----        6/01/2022   8:46 AM     16 hello.txt
```

Filter Pipeline using Where-Object

- Operators include -eq, -gt, -le, -ne, -not, -or, -and
  - E.g. list files of length greater or equal to 5 bytes

```
PS> ls | Where-Object Length -ge 5

    Directory: C:\Users\shield
Mode   Length Name
----   ------ ----
-a----     16 hello.txt
```

- List files that are either <20 bytes
  - PS> ls | Where-Object Length -lt 20

```
    Directory: C:\Users\shieldj
Mode   Length Name
----   ------ ----
d-r--         Downloads
-a---       7 hello.txt
```

Filter Pipeline using Where-Object

- List the connected network links
  - PS> **Get-NetAdapter** | **Where-Object** Status **-eq** Up

```
Name                                      Status LinkSpeed
----                                      ------ ---------
vEthernet (Default Switch)                Up     10 Gbps
Ethernet                                  Up     1 Gbps
```

Filter Pipeline using Select-Object

- Select head of pipeline (or tail with -last)

```
PS> ls | Select-Object -first 1
    Directory: C:\Users\shield
 Mode                LastWriteTime         Length Name
 ----                -------------         ------ ----
 d-----        11/03/2021  12:48 PM                Documents
```

- Select or rearrange columns

```
PS> ls | Select-Object Mode, Name -first 1
 Mode          Name
 ----          ----
 d-----        Documents
```

Filter Pipeline using Where-Object

- When was hello.txt's parent directory created?
  - FileInfo has Directory property

```
PS> ls hello* | Get-Member

   TypeName: System.IO.FileInfo
Name                        MemberType     Definition
Directory                   Property       System.IO.DirectoryInfo
Directory…
```

  - DirectoryInfo has CreationTime property

```
PS> (ls hello*).Directory | Get-Member
   TypeName: System.IO.DirectoryInfo
CreationTime                Property       datetime CreationTime…
```

```
PS> (ls hello*).Directory.CreationTime

Wednesday, 3 February 2021 8:43:31 AM
```

Variables

- Collection of objects can be stored

```
PS> $ext=ls hello.*
PS> $ext.Extension
.txt
```

References

Happy Studying!

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